

Coordinated Planning Strategy for Airborne Wind Energy in Ground-based Power Grids

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Abstract—High-altitude wind energy (300–3000 m) offers a promising renewable resource, particularly for inland regions with limited surface wind. Airborne Wind Energy Systems (AWESs) are capable of capturing and converting high-altitude wind into electricity, which provides a novel approach to diversify and decarbonize the energy supply. Unlike conventional wind systems, AWES can dynamically adjust their operating height to maximize energy capture. This flexibility, however, complicates the use of traditional energy coupling models, resulting in a relative lack of research on AWES integration into power systems. To address this, we propose a long-term planning and operational optimization framework for AWESs within multi-energy power systems. First, a two-stage stochastic programming model is developed to incorporate uncertain operating scenarios, while a mixed-integer linear formulation defines the AWES output potential. Second, to mitigate computational complexity, a problem-driven scenario reduction strategy based on the *PCK-Medoids* algorithm is applied. Finally, case studies across five western Chinese provinces demonstrate that AWES integration enhances clean energy complementarity and system performance: renewable penetration increases by up to 23 percentage points, storage requirements decline by 65%, and annualized costs fall by 14%. These findings highlight the potential of AWESs to reduce reliance on high-cost resources and support power system decarbonization.

Index Terms—Airborne wind energy, high-altitude wind energy, variable renewable energy, multi-energy power system, stochastic programming, scenario reduction.

I. INTRODUCTION

The global transition towards carbon-neutral power systems is driving unprecedented deployment of variable renewable energy (VRE), especially onshore and offshore wind and large-scale photovoltaic generation [1]. These sources are essential for deep decarbonization, yet their inherent variability and limited dispatchability substantially heighten system flexibility requirements [2]. As VRE penetration increases, power systems

demand larger reserve margins, faster ramping capabilities, and enhanced coordination between storage, demand-side, and flexible generation resources[3]. If these flexibility needs are unmet, operating costs can escalate nonlinearly, and reliability risks may amplify under extreme weather events.

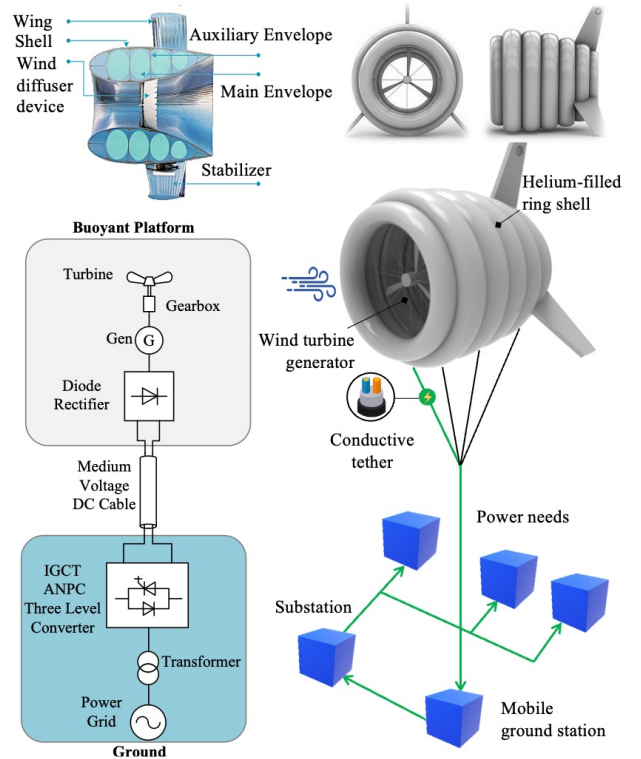


Fig. 1. Conceptual configuration of a buoyant airborne turbine. The platform integrates a wind turbine and generator within a helium-filled annular shell, which is tethered to a ground station for power transmission and operational control.

Airborne Wind Energy Systems (AWESs) have emerged as a promising technological pathway to enhance renewable integration by accessing stronger and more persistent wind resources at higher altitudes [4]. In this work, we focus on buoyant airborne platforms, as shown in Fig. 1, capable of sustained operation between altitudes of approximately 300 m and 3000 m. At these heights, wind speeds are generally stronger and more stable, and turbulence intensity is reduced compared to near-surface conditions [5]. A distinctive operational advantage of AWESs lies in their ability to adjust operating altitude to exploit favorable wind layers dynamically [6], thereby improving average power output, reducing

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short-term variability, and enhancing complementarity with other VRE resources such as photovoltaics. By exploiting high-quality wind resources while minimizing land-use requirements, AWESs offer a unique combination of improved generation performance and siting flexibility compared with conventional ground-based wind turbines (GB-WTs) [7].

Research on AWESs has advanced along several complementary lines of inquiry. From a theoretical perspective, aerodynamic and control studies have established the foundation for energy harvesting in buoyant airborne turbine (BAT) platforms. Samson [8] proposed a hierarchical control architecture for AWES operation, integrating Proportional–Integral–Derivative (PID) and Linear–Quadratic Regulator (LQR) controllers, model predictive control (MPC), and extremum-seeking algorithms to achieve flight stability and maximize power generation in real-time. Saleem and Kim [9] performed computational fluid dynamics (CFD) simulations to investigate the aerodynamic interaction between the buoyant envelope and the ducted wind turbine, highlighting the influence of envelope geometry and wake dynamics on overall performance.

On the technological front, prototype development and industrial demonstrations have validated the feasibility of AWESs. Altaeros Energies has deployed BAT-based systems capable of lifting turbines to altitudes above 300 m, delivering grid-compatible electricity and enabling rapid deployment in remote areas [10]. In China, a large-scale BAT prototype achieved a flight altitude exceeding 1000 m with more than 100 kW output in January 2025 [11]. At the same time, the Jixi High-Altitude Wind Power Demonstration Project successfully operated a buoyant system between 500 m and 3000 m, achieving megawatt-scale capacity [12].

Regarding atmospheric resource evaluation, reanalysis datasets and mesoscale simulations have been used to map high-altitude wind regimes relevant to BAT deployment. Winds between 300 m and several kilometers altitude exhibit higher speeds, lower turbulence, and reduced diurnal variability compared to near-surface resources [13], [14], offering potential improvements in both capacity factor and supply reliability. Despite these advances, system-level analyses of BAT integration remain scarce. Most techno-economic studies focus on isolated prototypes or assume altitude-invariant generation profiles, which can misrepresent operational flexibility, dispatchability, and interactions with the power systems on the ground [15]. A rigorous system-level modeling framework that explicitly incorporates altitude-dependent resource profiles, operational control strategies, and interactions with other grid assets is still lacking.

In parallel, the power system planning literature has developed advanced stochastic optimization frameworks to co-optimize long-term investments and short-term operations under uncertainty. Two-stage stochastic programming has been widely adopted for renewable-rich capacity expansion problems, where first-stage decisions determine asset investments and second-stage decisions model operational dispatch across representative scenarios [16]. The accuracy and tractability of such models depend critically on the size and representativeness of the scenario set: An excessive number of scenarios leads to prohibitive computational burdens, whereas poorly

selected scenarios can result in suboptimal or even misleading investment outcomes. Scenario reduction techniques, including probability-metric methods [17], [18] and clustering-based compression [19], have been developed to address this challenge. However, most existing approaches focus primarily on statistical representativeness and often overlook the operational and economic sensitivities of the specific planning problem, which can cause the reduced scenario set to lose decision-critical variability.

Motivated by this gap, this paper is the first to propose an integrated stochastic system planning framework for *air-ground co-optimization*, achieving coordinated deployment of high-altitude AWESs and ground-based storage resources. The proposed framework incorporates detailed AWES operational characteristics, explicitly modeling altitude-varying wind resources and their interactions with other generation and storage assets. To preserve investment-relevant variability while ensuring tractability, we further design a novel *Problem-based Chance-Cost K-Medoids (PCCCK-Medoids)* scenario reduction method that extracts high-quality representative scenarios aligned with the economic sensitivities of planning decisions.

The main contributions of this work are as follows:

- 1) We present the first quantitative assessment of AWES deployment in a realistic multi-province, high-VRE system. By capturing the complementarity between high-altitude winds and regional renewable profiles, this study highlights the unique system-level benefits of AWESs compared with conventional resource assessments.
- 2) We develop a stochastic planning framework that embeds AWES operational envelopes into a two-stage capacity expansion model. The framework explicitly represents altitude-varying wind profiles, altitude adjustment costs, and interactions with conventional units, PV, and storage, thereby enabling a holistic evaluation of AWESs within integrated multi-energy systems.
- 3) We introduce methodological advances to support tractable large-scale applications. A mix-integer non-linear programming (MINLP) altitude scheduling model is proposed and further linearized into a mix-integer linear programming (MILP) form to accelerate simulations. In addition, a problem-driven *PCCCK-Medoids* scenario reduction method is designed to preserve decision-critical variability and deliver investment-sensitive representative scenarios for planning.

II. MODELLING OF AIRBORNE WIND ENERGY SYSTEMS

This section models the generation potential of AWESs and GB-WTs using hourly ECMWF fifth-generation (ERA5) reanalysis wind data and system-specific power curves. For each time step, the maximum feasible output is derived as an upper bound on generation, which is then imposed in the system optimization model as a capacity constraint.

A. Altitude Optimization Model

To quantify the energy output of AWESs, we model the altitude selection of AWESs as the following optimization

problem with a total of T time steps. For $i \in [0, N-1]$ and $t \in [1, T]$, we have

$$\max_{H_t, z_t, r_{i,t}} \sum_{t=1}^T (P_t^{\text{gen}} - P_t^{\text{mai}} - P_t^{\text{cha}}) \Delta t, \quad (1)$$

$$\text{s.t. } P_t^{\text{gen}} = k_1 \rho_t \min\{W_t, W_{\text{rated}}\}^3, \quad (2)$$

$$P_t^{\text{mai}} = k_2 \rho_t W_t^2, \quad (3)$$

$$P_t^{\text{cha}} = k_3 W_t^2 |\dot{H}_t|, \quad (4)$$

$$r_{i,t} h_i \leq H_t < r_{i,t} h_{i+1} + (1 - r_{i,t}) K, \quad (5)$$

$$\sum_{i=0}^{N-1} r_{i,t} = 1, \quad (6)$$

$$H_0 = h_0, \quad (7)$$

$$|H_t - H_{t-1}| \leq S_{\text{max}} \Delta t, \quad (8)$$

$$0 \leq |H_t - H_{t-1}| \leq z_t K, \quad (9)$$

$$\sum_{t=s}^{s+D} z_t \leq 1, \quad (10)$$

$$W_t = \sum_{i=0}^{N-1} r_{i,t} \left[W_{h_i} + \frac{W_{h_{i+1}} - W_{h_i}}{h_{i+1} - h_i} (H_t - h_i) \right], \quad (11)$$

$$\rho_t = \sum_{i=0}^{N-1} r_{i,t} \left[\rho_{h_i} + \frac{\rho_{h_{i+1}} - \rho_{h_i}}{h_{i+1} - h_i} (H_t - h_i) \right]. \quad (12)$$

At each time step t , the decision variables comprise: the operational altitude H_t , the binary indicator z_t , and the binary interval selectors $\{r_{i,t}\}_{i=0}^{N-1}$. Here, z_t signals altitude adjustment occurrence, while $r_{i,t}$ indicates whether H_t lies within the altitude interval $[h_i, h_{i+1})$. The objective function (1) maximizes the net energy output by considering power generation P_t^{gen} , maintenance power P_t^{mai} , and the power required for altitude adjustments P_t^{cha} at each time step. Specifically, the AWES's generated power P_t^{gen} is computed via equation (2) using the actual wind speed W_t , rated wind speed W_{rated} , and air density ρ_t . Taking the minimum value means that the power generated by the AWESs will be limited to the rated value when the wind speed is too high. As formalized in equations (3)-(4), the term P_t^{mai} represents the power required to maintain the AWESs at a fixed altitude, while P_t^{cha} quantifies the power consumption during altitude transitions, with $|\dot{H}_t|$ denoting the rate of altitude change. The coefficients k_1 , k_2 , and k_3 are empirically derived from experimental studies [20].

In the actual operation of AWESs, constraints on altitude states and altitude changes are of paramount importance [21]. Regarding altitude states, constraint (5)-(6) collectively mandate that at time step t , the AWESs must operate exclusively within one altitude interval $[h_i, h_{i+1})$. This set of intervals is defined by the partition $H_{\text{lb}} = h_0 < h_1 < \dots < h_N = H_{\text{ub}}$, where $[H_{\text{lb}}, H_{\text{ub}}]$ denotes the optimal operational altitude range and $h_{i+1} - h_i \equiv \Delta h$ for each i . Concerning altitude changes, constraint (8) restricts the altitude variation per unit time by a speed limit S_{max} , whereas constraints (9)-(10) stipulate that the AWESs may perform at most one altitude adjustment during any consecutive D time steps.

In practical applications, the data of wind speed W_t and air density ρ_t are only available at discrete altitude points,

i.e., $h_0, \dots, h_N$. To obtain the data at arbitrary altitudes for model input, we employ a linear interpolation technique to fill these data gaps [22]. For any altitude h , when it belongs to the interval $[h_i, h_{i+1})$, the wind speed and air density at that altitude are calculated as follows:

$$W(h) = W_{h_i} + \frac{W_{h_{i+1}} - W_{h_i}}{h_{i+1} - h_i} (h - h_i), \quad (13)$$

$$\rho(h) = \rho_{h_i} + \frac{\rho_{h_{i+1}} - \rho_{h_i}}{h_{i+1} - h_i} (h - h_i). \quad (14)$$

Thus, constraints (11)-(12) specify the wind speed and air density parameters corresponding to the altitude variable H_t when it falls within a specific altitude interval.

B. Linearly Reformulated Model

The model presented in II-A is a MINLP problem, for which we make some simplifications as well as linear approximations in order to enhance computational efficiency. We discretize the altitude decision variable H_t into a finite set of predefined values by selecting M representative levels $h_i^{(m)}$ within each altitude interval $[h_i, h_{i+1})$. This approximation covers the entire interval, with $h_i^{(m+1)} - h_i^{(m)} \equiv \Delta h/M$ holding uniformly for all i and m . By defining the binary variable $r_{i,t}^{(m)}$, the decision variable H_t can be approximated the as

$$H_t \approx \bar{H}_t = \sum_{i=0}^{N-1} \sum_{m=0}^{M-1} r_{i,t}^{(m)} h_i^{(m)}. \quad (15)$$

The data of wind speed $W_i^{(m)}$ and air density $\rho_i^{(m)}$ at each level $h_i^{(m)}$ can be obtained by equations (13)-(14). By defining the notation

$$\bar{W}_t^2 := \sum_{i=0}^{N-1} \sum_{m=0}^{M-1} r_{i,t}^{(m)} (W_i^{(m)})^2, \quad (16)$$

$$W_t^s := \sum_{i=0}^{N-1} \sum_{m=0}^{M-1} r_{i,t}^{(m)} \rho_i^{(m)} (W_i^{(m)})^2, \quad (17)$$

$$W_t^c := \sum_{i=0}^{N-1} \sum_{m=0}^{M-1} r_{i,t}^{(m)} \rho_i^{(m)} \min\{W_i^{(m)}, W_{\text{rated}}\}^3, \quad (18)$$

constraints (2)-(3) can be reformulated as

$$P_t^{\text{gen}} \approx \bar{P}_t^{\text{gen}} = k_1 W_t^c, \quad (19)$$

$$P_t^{\text{mai}} \approx \bar{P}_t^{\text{mai}} = k_2 W_t^s, \quad (20)$$

The rate of altitude change $\dot{H}_t$ in constraint (4) is approximated as the average rate of change during the time interval Δt between consecutive time steps. Consequently, for $t \in [1, T]$, P_t^{cha} is approximated as

$$P_t^{\text{cha}} \approx \bar{P}_t^{\text{cha}} = k_3 \bar{W}_t^2 |\bar{H}_t - \bar{H}_{t-1}| / \Delta t, \quad (21)$$

with the initial condition $\bar{H}_0 = h_0$. However, since both variables $\bar{W}_t^2$ and $\bar{H}_t$ are linear combinations of binary variables, making their product a nonlinear term, we introduce an auxiliary binary variable $r_{i,j,t}^{(m,n)}$ to handle it. For $i, j \in [0, N-1]$, $m, n \in [0, M-1]$, and $t \in [2, T]$, define

$$r_{i,j,t}^{(m,n)} = r_{i,t}^{(m)} \cdot r_{j,t-1}^{(n)}. \quad (22)$$

Then for $t \in [2, T]$, $\bar{P}_t^{\text{cha}}$ can be expressed as

$$\bar{P}_t^{\text{cha}} = \sum_{i,j=0}^{N-1} \sum_{m,n=0}^{M-1} r_{i,j,t}^{(m,n)} (W_i^{(m)})^2 |h_i^{(m)} - h_j^{(n)}| / \Delta t, \quad (23)$$

and for the initial time step $t = 1$, this simplifies to

$$\bar{P}_1^{\text{cha}} = \sum_{i=0}^{N-1} \sum_{m=0}^{M-1} r_{i,t}^{(m)} (W_i^{(m)})^2 h_i^{(m)} / \Delta t. \quad (24)$$

After simplification, the model reduces to an MILP problem. Specifically, for $i, j \in [0, N-1]$, $m, n \in [0, M-1]$, and $t \in [1, T]$, we have

$$\max_{z_t, r_{i,t}^{(m)}, r_{j,t}^{(m,n)}} \sum_{t=1}^T (\bar{P}_t^{\text{gen}} - \bar{P}_t^{\text{mai}} - \bar{P}_t^{\text{cha}}) \Delta t, \quad (25)$$

$$\text{s.t.} \quad \sum_{i=0}^{N-1} \sum_{m=0}^{M-1} r_{i,t}^{(m)} = 1, \quad (26)$$

$$\bar{H}_0 = h_0, \quad (27)$$

$$0 \leq |\bar{H}_t - \bar{H}_{t-1}| \leq \min\{S_{\max} \Delta t, z_t K\} \quad (28)$$

$$r_{i,j,t}^{(m,n)} \leq \min\{r_{i,t}^{(m)}, r_{j,t}^{(n)}\}, \quad \forall i, j, m, n, \quad (29)$$

$$r_{i,j,t}^{(m,n)} \geq r_{i,t}^{(m)} + r_{j,t}^{(n)} - 1, \quad \forall i, j, m, n, \quad (30)$$

$$(10), (19), (20), (23), (24).$$

III. CAPACITY EXPANSION MODEL WITH AWES INTEGRATION

This section formulates the capacity planning problem for an integrated power system comprising wind power, solar energy, and energy storage systems (BESS), considering uncertainty in renewable generation and power demand. We compare the advantages of AWESs with GB-WTs for wind power generation investments.

A. Stochastic Programming Framework

1) *Structure of the Two-Stage Model:* We adopt a two-stage stochastic programming framework: The first stage determines the investment decisions for renewable energy and storage assets, while the second stage evaluates the expected optimal operational cost under multiple scenarios, conditioned on the established portfolio. The investment cost is defined as:

$$IC(x) = \sum_{i=1}^R (x_i^w C_i^w + x_i^b C_i^b), \quad (31)$$

where x_i^w, x_i^b are the installed capacities of wind turbines and BESS at bus i , with C_i^w, C_i^b representing the corresponding per-unit capacity costs. The decision variables across R sites are collectively denoted as $x = ((x_i^w, x_i^b))_{i=1}^R$. In the second stage, for each scenario ξ , the operational cost over T periods is:

$$OC(y; \xi) = \sum_{t=1}^T \sum_{i=1}^R (C_i^g p_{i,t}^g(\xi) + C_i^{\text{sh}} l_{i,t}^{\text{sh}}(\xi)), \quad (32)$$

where $p_{i,t}^g(\xi)$ denotes the thermal power generation and $l_{i,t}^{\text{sh}}(\xi)$ denotes the load shedding at bus i and time t . The decision

variables for scenario ξ are collectively denoted as $y_\xi = ((p_{i,t}^g(\xi), l_{i,t}^{\text{sh}}(\xi)))_{i,t=1}^{R,T}$.

Our ultimate goal is to minimize the sum of investment cost and operational cost, formally expressed as:

$$\min_{x \in \mathcal{X}} TC(x, \Xi) = IC(x) + \mathbb{E}_\xi \left[\min_{y \in \mathcal{Y}(x, \xi)} OC(y; \xi) \right]. \quad (33)$$

Here, Ξ denotes the set of all scenarios with associated probability distribution F . The set $\mathcal{X}$ is the feasible set of device capacities, and the set $\mathcal{Y}(x, \xi)$ encodes all operational constraints based on the first-stage decisions x and the scenario realization ξ . For ease of subsequent presentation, we define the function G as

$$G(y, \xi | x) = \min_{y \in \mathcal{Y}(x, \xi)} OC(y; \xi), \quad (34)$$

and denote the optimal solution under scenario ξ_j and ζ_k as $y_{\xi_j}^*$ and $y_{\zeta_k}^*$, respectively.

2) *Scenario Generation and Reduction Method:* In practice, the distribution of renewable generation intensity and regional load are unknown, but an empirical probability distribution $\mathbb{P} = \{p_j\}_{j=1}^J$ can be constructed by independently sampling a finite set $\Xi_\xi = \{\xi_j\}_{j=1}^J$ of scenarios from historical data [23]. To ensure sufficient accuracy, the value of J is typically set large enough. Then the total cost can be expressed as

$$TC(x, \Xi_\xi) = IC(x) + \sum_{j=1}^J p_j G(y_{\xi_j}^*, \xi_j | x). \quad (35)$$

To ensure tractability for large J , we perform scenario reduction to obtain a smaller set $\Xi_\zeta = \{\zeta_k\}_{k=1}^K$ derived from Ξ_ξ , with associated probabilities q_k , yielding:

$$TC(x, \Xi_\zeta) = IC(x) + \sum_{k=1}^K q_k G(y_{\zeta_k}^*, \zeta_k | x). \quad (36)$$

Our objective is to construct a proper distribution $\mathbb{Q}$ such that $TC(x, \Xi_\zeta)$ closely approximates the empirical total cost $TC(x, \Xi_\xi)$, while substantially reducing the computational burden. Details of the proposed problem-based chance-cost K -medoids reduction algorithm and error analysis are presented in Section IV.

B. Power System Constraints

This section introduces operational constraints for the power system, where lowercase letters denote decision variables and uppercase letters represent known or intermediate variables. Under the investment x and the scenario ξ , each constraint holds for all sites $i = 1, \dots, R$ and time periods $t = 1, \dots, T$.

1) *Power Balance Constraints:* These constraints ensure supply-demand balance in power systems.

$$\begin{aligned} & p_{i,t}^g(\xi) + p_{i,t}^{\text{PV}}(\xi) + p_{i,t}^{\text{WT}}(\xi) + p_{i,t}^{\text{dis}}(\xi) - p_{i,t}^{\text{ch}}(\xi) \\ & + \sum_{l=1}^{N_i^{\text{in}}} f_{l,t}^{\text{in}}(\xi) - \sum_{l=1}^{N_i^{\text{out}}} f_{l,t}^{\text{out}}(\xi) = D_{i,t}(\xi) - l_{i,t}^{\text{sh}}(\xi), \end{aligned} \quad (37)$$

where $p_{i,t}^{\text{PV}}(\xi)$ and $p_{i,t}^{\text{WT}}(\xi)$ represent the photovoltaic and wind power generation, $p_{i,t}^{\text{ch}}(\xi)$ and $p_{i,t}^{\text{dis}}(\xi)$ denote the BESS charging and discharging power, $f_{l,t}^{\text{in}}(\xi)$ and $f_{l,t}^{\text{out}}(\xi)$ are the incoming

and outgoing power flow on line l , N_i^{in} and N_i^{out} indicate the number of incoming and outgoing lines at bus i , and $D_{i,t}(\xi)$ is the total load.

2) *Generation Capacity Limits*: These constraints establish bounds on power generation capabilities. Since we assume no costs for wind and solar curtailment, wind and solar generation are treated as decision variables.

$$0 \leq p_{i,t}^g(\xi) \leq P_{i,t}^{g,\max}, \quad (38)$$

$$0 \leq p_{i,t}^{\text{PV}}(\xi) \leq P_{i,t}^{\text{PV},\max}(\xi), \quad (39)$$

$$0 \leq p_{i,t}^{\text{WT}}(\xi) \leq P_{i,t}^{\text{WT},\max}(\xi), \quad (40)$$

where $P_{i,t}^{g,\max}$ represent the generation upper bounds of thermal generators at bus i , and $P_{i,t}^{\text{PV},\max}(\xi)$ and $P_{i,t}^{\text{WT},\max}(\xi)$ denote the upper bounds on photovoltaic and wind generation.

The power generation upper bound $P_{i,t}^{\text{WT},\max}(\xi)$ is formulated as a decision variable because, for AWESs, BAT's altitude can be dynamically adjusted, rendering the power upper bound curve unknown even under a specified scenario. The power upper bound at each time step is defined using the computation results by the MILP model in Section II-B, which maximizes the power output over T consecutive time intervals. Specifically, the upper bound $P_{i,t}^{\text{WT},\max}(\xi)$ becomes

$$P_{i,t}^{\text{AWES},\max}(\xi) = \bar{P}_{i,t}^{\text{gen}}(\xi) - \bar{P}_{i,t}^{\text{mai}}(\xi) - \bar{P}_{i,t}^{\text{cha}}(\xi), \quad (41)$$

where $\bar{P}_{i,t}^{\text{gen}}(\xi)$, $\bar{P}_{i,t}^{\text{mai}}(\xi)$, and $\bar{P}_{i,t}^{\text{cha}}(\xi)$ denote the optimized power output, the power consumption for maintaining the AWES's attitude, and the power consumption for altitude adjustments, respectively.

As a comparison, conventional GB-WTs operate at a fixed altitude (typically 200 meters above ground), resulting in a deterministic generation upper bound curve. Then the variable $P_{i,t}^{\text{WT},\max}(\xi)$ becomes

$$P_{i,t}^{\text{GBWT},\max}(\xi) = k_1 \rho \min \{W_{i,t}(\xi), W_{\text{rated}}^{\text{GBWT}}\}^3, \quad (42)$$

where ρ is the air density, $W_{i,t}(\xi)$ is the wind speed, and $W_{\text{rated}}^{\text{GBWT}}$ is the rated wind speed.

3) *Load Shedding Constraints*: This constraint ensures that load shedding must be non-negative and no greater than the total demand.

$$0 \leq L_{b,t}^{\text{sh}}(\xi) \leq D_{b,t}(\xi). \quad (43)$$

4) *DC Power Flow Constraints*: Flow constraints on each transmission line l are modeled using a linear approximation.

$$f_{l,t}(\xi) = B_l(\theta_{l(s),t}(\xi) - \theta_{l(r),t}(\xi)), \quad (44)$$

$$0 \leq f_{l,t}(\xi) \leq P_l^{\text{line},\max}, \quad (45)$$

where B_l is the susceptance of line l , $\theta_{l(s),t}(\xi)$ and $\theta_{l(r),t}(\xi)$ are the voltage phase angles at the sending and receiving buses of line l , and $P_l^{\text{line},\max}$ represents the thermal loading limit of line l .

5) *State-of-Charge Dynamics*: The change in state of charge is constrained by the following relationship.

$$E_{i,t}(\xi) = E_{i,t-1}(\xi) + (\eta^{\text{ch}} p_{i,t}^{\text{ch}}(\xi) - \frac{1}{\eta^{\text{dis}}} p_{i,t}^{\text{dis}}(\xi)) \Delta t, \quad (46)$$

$$E_{i,0}(\xi) = E_i^0, \quad (47)$$

$$0 \leq E_{i,t}(\xi) \leq x_i^b, \quad (48)$$

where $E_{i,t}(\xi)$ denotes the energy stored in the BESS with initial level E_i^0 , while η^{ch} and η^{dis} represent the charging and discharging efficiencies.

6) *Charging and Discharging Constraints*: Charging and discharging operations are mutually exclusive and subject to maximum power constraints.

$$0 \leq p_{i,t}^{\text{ch}}(\xi) \leq u_{i,t}^{\text{ch}}(\xi) P_i^{\text{ch},\max}, \quad (49)$$

$$0 \leq p_{i,t}^{\text{dis}}(\xi) \leq u_{i,t}^{\text{dis}}(\xi) P_i^{\text{dis},\max}, \quad (50)$$

$$u_{i,t}^{\text{ch}}(\xi) + u_{i,t}^{\text{dis}}(\xi) \leq 1, \quad (51)$$

where $u_{i,t}^{\text{ch}}(\xi)$ and $u_{i,t}^{\text{dis}}(\xi)$ are binary variables, while $P_i^{\text{ch},\max}$ and $P_i^{\text{dis},\max}$ represent the maximum instantaneous charging and discharging power.

IV. PROBLEM-BASED SCENARIO REDUCTION

We propose a problem-based scenario reduction method. In contrast to traditional scenario clustering via measuring statistical distances between the scenarios themselves, our scheme defines a novel metric and performs scenario reduction under this distance measure. Furthermore, we prove that the total cost associated with the reduced scenario set exhibits an error upper bound relative to the original set, demonstrating the reduction's validity.

A. Chance-Cost Distance

To establish a metric for total cost variations across different scenarios, we first define the total cost function for a given scenario ξ as

$$TC(x, \xi) = IC(x) + G(y_\xi^*, \xi | x), \quad (52)$$

and denote the scenario-dependent optimal solution minimizing this total cost by

$$x_\xi^* = \arg \min_{x \in \mathcal{X}} TC(x, \xi). \quad (53)$$

Then, we define the distance between two scenarios ξ_i and ξ_j as the sum of the additional total cost incurred when each scenario's optimal solution, $x_{\xi_i}^*$ or $x_{\xi_j}^*$, is applied to the other, and refer to this measure as the *chance-cost* distance.

Definition 1 (Chance-Cost Distance). *The distance between two scenarios ξ_i and ξ_j is*

$$d(\xi_i, \xi_j) := \left(TC(x_{\xi_j}^*, \xi_i) - TC(x_{\xi_i}^*, \xi_i) \right) + \left(TC(x_{\xi_i}^*, \xi_j) - TC(x_{\xi_j}^*, \xi_j) \right). \quad (54)$$

Lemma 1 (Properties). *The distance $d(\xi_i, \xi_j)$ satisfies:*

- 1) **Consistency**: $d(\xi_i, \xi_j) = 0$ iff $\xi_i = \xi_j$ and $x_{\xi_i}^* = x_{\xi_j}^*$.
- 2) **Symmetry**: $d(\xi_i, \xi_j) = d(\xi_j, \xi_i)$.
- 3) **Convergence**: $\sup \{d(\xi_i, \xi_j) : \xi_i, \xi_j \in \Xi_\xi, \|\xi_i - \xi_j\| \leq \delta\}$ tends to 0 as $\delta \rightarrow 0$.
- 4) **Bounded Growth**: There exists bounded function $\lambda : \Xi_\xi \rightarrow \mathbb{R}_+$ such that

$$d(\xi_i, \xi_j) \leq \lambda(\xi_i) + \lambda(\xi_j), \quad \forall \xi_i, \xi_j \in \Xi_\xi.$$

The proof can be found in [24].

B. Scenario Reduction via PCCK-Medoids

We develop Algorithm 1 to construct the set $\Xi_\zeta = \{\zeta_k\}_{k=1}^K$ from the full set $\Xi_\xi = \{\xi_j\}_{j=1}^J$ by solving a clustering problem based on the distance $d(\xi_i, \xi_j)$, where representative scenarios (medoids) are selected to minimize total cost distortion.

Algorithm 1 Chance-Cost-Based Scenario Reduction

Require: Scenario set $\Xi_\xi = \{\xi_j\}_{j=1}^J$ with distribution $\mathbb{P}$, and prescribed number of clusters K .

- 1: Find $x_{\xi_j}^*$ for each scenario ξ_j .
 - 2: Construct chance-cost distance matrix $D \in \mathbb{R}^{J \times J}$ with $D_{ij} := d(\xi_i, \xi_j)$.
 - 3: Apply K-medoids clustering to partition the scenarios into clusters $\{\mathcal{G}_k\}_{k=1}^K$.
 - 4: **for** each cluster $\mathcal{G}_k$ **do**
 - 5: Set medoid $\zeta_k := \arg \min_{\xi \in \mathcal{G}_k} \sum_{\xi' \in \mathcal{G}_k} d(\xi, \xi')$.
 - 6: Assign weight $q_k := \sum_{\xi_j \in \mathcal{G}_k} p_j$.
 - 7: **end for**
 - 8: **return** $\Xi_\zeta = \{\zeta_k\}_{k=1}^K$ with distribution $\{q_k\}_{k=1}^K$.
-

C. Error Upper Bound

In this section, we establish an upper bound on the difference between the total cost before and after scenario reduction, thereby demonstrating the validity of the proposed scenario reduction method.

Following the results in [24], for a function $TC(x, \xi)$ that is locally Lipschitz continuous and a distance metric $d(\xi_i, \xi_j)$ satisfying Lemma 1, it holds that for any $x \in \mathcal{X}$ and $\xi_i, \xi_j \in \Xi_\xi$,

$$|TC(x, \xi_i) - TC(x, \xi_j)| \leq L(\|x\|) \cdot d(\xi_i, \xi_j), \quad (55)$$

where L denotes a non-decreasing function. Let x^* and $\hat{x}^*$ denote the optimal solutions obtained by minimizing the objective functions given in (35) and (36), respectively. Then we have the following theorem.

Theorem 1 (Upper Bound). *Based on inequality (55), the error bound of the total costs before and after scenario reduction under the original scenario set is*

$$\begin{aligned} & TC(\hat{x}^*, \Xi_\xi) - TC(x^*, \Xi_\xi) \\ & \leq 2L_{\max} \sum_{k=1}^K \sum_{\xi_j \in \mathcal{G}_k} p_j d(\xi_j, \zeta_k), \end{aligned} \quad (56)$$

where $L_{\max} = \max\{L(\|\hat{x}^*\|), L(\|x^*\|)\}$.

The proof follows standard arguments in scenario approximation theory [25], [26].

V. CASE STUDIES

A. AWES Model and Parameters

As the core innovation of this study, AWES units are explicitly modeled with detailed physical and operational characteristics, as illustrated in Fig. 1. Each unit has a rated power of $P_{\text{rated}} = 100 \text{ kW}$ at a rated wind speed of $W_{\text{rated}} = 12 \text{ m/s}$. Aerodynamic performance is parameterized by three empirical

coefficients: $k_1 = 0.04825 \text{ kW} \cdot \text{s}^3 \cdot \text{kg}^{-1}$ for wind-to-power conversion efficiency, $k_2 = 0.075 \text{ kW} \cdot \text{m} \cdot \text{s}^2 \cdot \text{kg}^{-1}$ for sustaining altitude, and $k_3 = 0.125 \text{ kW} \cdot \text{s}^3 \cdot \text{kg}^{-1}$ for altitude transitions [20].

Operational limits specify an altitude range of $H_{\text{lb}} = 500 \text{ m}$ to $H_{\text{ub}} = 3000 \text{ m}$, with an initial altitude of $H_{\text{init}} = 500 \text{ m}$, a maximum altitude change rate of $S_{\text{max}} = 0.83 \text{ m/s}$, and a temporal resolution of $\Delta t = 1 \text{ h}$. A minimum dwell time $D = 1 \text{ h}$ is further imposed to ensure operational stability.

B. Power System and Data Description

1) *System structure and load data:* The case study is based on the HRP-38 test system, a 750 kV transmission network with 38 buses and 192 branches across five regions (D1–D5), as shown in Fig. 2. It applies coordinated multi-energy dispatch integrating thermal units, PV, wind power, and energy storage. Thermal and PV capacities, load levels, and hourly demand profiles are adopted from [27], which provides representative values for peak load, annual demand, and regional load distribution.

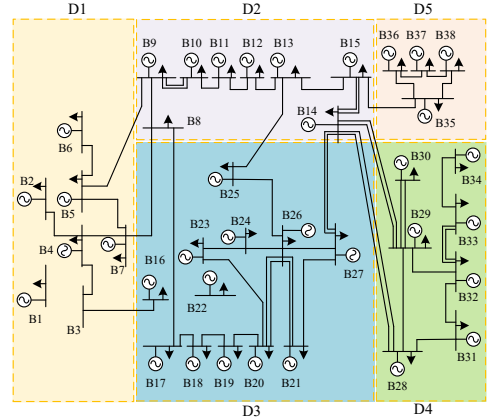


Fig. 2. Topology of the HRP-38 system [27].

2) *Renewable energy data:* Renewable generation includes PV, GB-WTs, and AWESs. PV profiles are sourced from [27]. Wind outputs for GB-WTs and AWESs are derived from ERA5 reanalysis data using a standard 100 kW turbine model. GB-WT parameters are calibrated with measured wind speeds from a representative Chinese province in 2023, achieving 2512 annual utilization hours [28]. AWES generation is modeled by a dynamic altitude optimization strategy that maximizes cumulative power over the scheduling horizon T .

3) *Investment and operational parameters:* Economic parameters are summarized in Table I. Capital expenditures are annualized using the equivalent annual cost method with a discount rate of 7%. The service life is assumed to be 20 years for both GB-WTs and AWESs, and 10 years for the BESS, which operates with a round-trip efficiency of 90%. In addition, the operational cost of PV generation is assumed to be negligible.

C. Model and Method Validation

Two validation procedures are implemented to demonstrate the robustness of the proposed framework. First, the accuracy

TABLE I
INVESTMENT AND OPERATIONAL COSTS

Notation	Description	Value
C_i^w	Capital cost of GB-WT	7000 CNY/kW
C_i^b	Capital cost of AWES	15000 CNY/kW
C_i^e	Capital cost of energy storage	795 CNY/kWh
C_i^g	Operational cost of thermal generation	0.3 CNY/kWh

of the linearized optimization model is evaluated against the original nonlinear formulation in section II-B. Second, the effectiveness of the proposed cost-oriented scenario reduction method is compared against a conventional distribution-driven clustering approach in section IV-B.

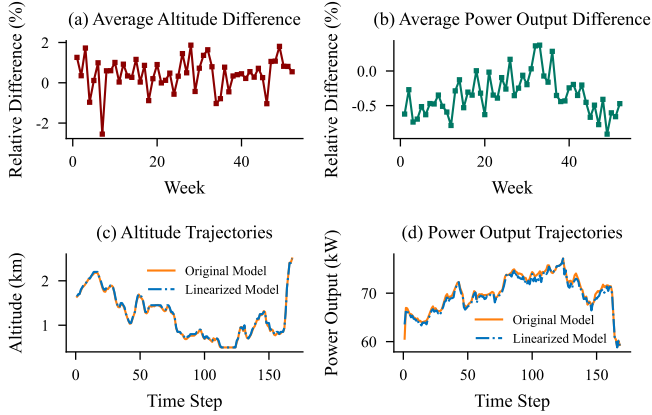


Fig. 3. Comparison of the original and linearized models in terms of altitude decisions and power output.

1) *Validation of the Linearized Optimization Model:* To improve computational tractability, the nonlinear AWES model is discretized and linearized. The validity of this approximation is evaluated using hourly wind data over a 7-day horizon ($T = 168$ time steps) across 52 datasets, where each dataset corresponds to one consecutive week of wind speed records, collectively covering an entire year.

As shown in Fig. 3(a)-(b), the original nonlinear formulation yields an average power output of 49.64 kW, compared with 49.66 kW from the linearized model, corresponding to a negligible relative error of 0.04%. The average operating altitude is 1206 m for the nonlinear formulation and 1219 m for the linearized model, resulting in a relative error of 1.08%. Furthermore, Fig. 3(c)-(d) illustrates the results for a representative week, where the linearized trajectories exhibit nearly identical trends to the nonlinear benchmark in both altitude decisions and power generation.

In terms of computational efficiency, the linearized model achieves an average solution time of 12.4 s per instance, compared to 158.7 s for the original formulation, representing a speed-up factor of approximately 12.8 times.

These results confirm that the linearized model maintains high fidelity in both decision variables and objective values while significantly reducing computational effort, making it suitable for large-scale stochastic planning problems.

2) *Validation of the Scenario Reduction Method:* The proposed *PCK-Medoids* method is validated through a two-stage

procedure. First, reduced scenarios are applied to the planning model to obtain investment decisions. Second, these decisions are evaluated on the original uncertainty set to compute the total system cost. A lower total cost indicates that the reduced scenarios better preserve decision-relevant uncertainty. As a benchmark, we adopt the distribution-based K-means method, which is widely used in scenario reduction but relies only on probabilistic similarity.

TABLE II
COMPARISON OF DIFFERENT CLUSTERING METHODS

Configuration	k	<i>PCK-Medoids</i> (Billion CNY)	K-means (Billion CNY)	Relative Difference (%)
AWES	5	311.45	301.28	-3.27
	6	311.61	300.77	-3.48
	7	311.30	299.81	-3.69
	8	295.12	299.67	1.54
	9	295.12	299.64	1.54
GB-WT	5	616.04	655.64	6.43
	6	530.49	534.67	0.79
	7	530.62	534.61	0.75
	8	530.47	532.38	0.36
	9	529.57	536.39	1.29

As shown in Table II, *PCK-Medoids* achieves lower or comparable costs in most cases. For the AWES configuration, it shows clear advantages at $k = 8$ and $k = 9$, while the slight underperformance at $k = 7$ reflects a local imbalance in scenario representation. For the GB-WT configuration, it yields significant savings under small k (e.g., 6.43% at $k = 5$) and remains competitive as k increases. These findings demonstrate that cost-aware clustering effectively identifies representative weeks.

Based on the clustering analysis, eight clusters ($k = 8$) are identified for both the AWES and GB-WT storage systems. The representative weeks for AWES are weeks 8, 10, 13, 19, 25, 38, 46, and 50, whereas for GB-WT, they are weeks 2, 6, 11, 19, 21, 35, 46, and 50.

D. Capacity Planning Results

To assess the impact of wind technology choice on system expansion, we apply the proposed model under two scenarios: the GB-WT configuration, integrating GB-WTs, PV, thermal units, and BESS; and the AWES configuration, where GB-WTs are replaced by AWESs while PV and thermal capacities are kept identical to ensure comparability.

TABLE III
PLANNING RESULTS FOR GB-WT AND AWES CONFIGURATIONS

Metric	AWES	GB-WT
Installed WT capacity x_i^w	243 GW	200 GW
Installed BESS capacity x_i^b	59 GW	168 GW
Storage-to-wind ratio	24 %	84 %

As summarized in Table III, replacing GB-WTs with AWESs substantially reduces BESS requirements from 168 GW to 59 GW, representing a 65% reduction, while

requiring a moderate increase of 22% in wind capacity. Consequently, the wind-to-storage capacity ratio decreases from 84% in the GB-WT configuration to 24% in the AWES configuration, indicating that AWESs can operate with significantly lower dependence on costly storage infrastructure.

These capacity outcomes highlight the reduced storage dependence enabled by AWES deployment. To further examine the implications at the operational level, we next analyze the generation mix and short-term system performance.

E. Generation Mix Analysis

In terms of the annual generation mix, the AWES configuration increases the share of clean energy from 52.9% to 75.7%, as shown in Fig. 4. This improvement is primarily driven by a 60% increase in wind generation, from 37.3% to 59.8%, which enables substantial displacement of thermal generation. These results demonstrate the potential of AWESs to accelerate decarbonization and enhance system sustainability.

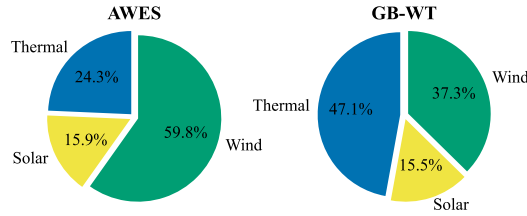


Fig. 4. Annual generation mix under GB-WT and AWES configurations.

F. Operational Performance Analysis

While the planning results demonstrate the structural advantages of AWES adoption, it is also important to evaluate operational performance. This section compares the GB-WT and AWES configurations over both a representative 168-hour week, highlighting detailed generation profiles, and the aggregated statistics of representative scenarios, with focus on thermal flexibility, renewable variability and BESS dynamics.

1) *Overall Power Output and Thermal Flexibility*: Fig. 5 illustrates the hourly generation profiles. The outer curve indicates that the AA-AWES system provides a smoother output, while the shaded area shows a significant reduction in BESS charging and discharging activities.

Table IV further quantifies these effects. AWES integration reduces the thermal peak-to-valley range by 16.7% and raises the minimum output ratio by 17.5 percentage points, indicating improved thermal utilization. Although ramping frequency rises slightly, ramp magnitudes are lower, suggesting finer and less disruptive dispatch adjustments. Overall, AWES improves operational stability while alleviating storage requirements.

2) *Renewable Variability and PV Synergy*: To evaluate the operational benefits of AWES deployment, we examine both the variability of wind output and its temporal correlation with PV generation. As shown in Table V, AWES significantly enhances wind generation stability, increasing the mean output by 54.8% and reducing variability by 40.3%. This smoother output reduces the need for fast-ramping reserves and alleviates storage requirements.

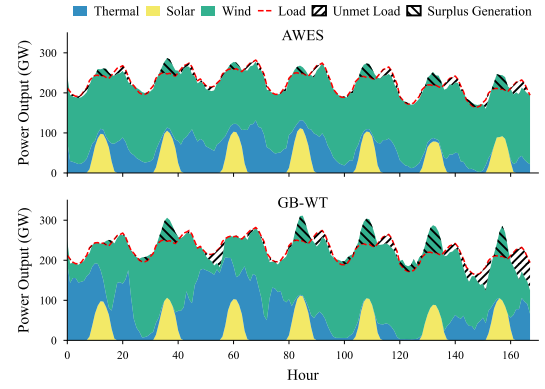


Fig. 5. Hourly generation profiles under AWES and GB-WT configurations.

TABLE IV
THERMAL DISPATCH VARIABILITY METRICS UNDER AWES AND GB-WT CONFIGURATIONS.

Metric	AWES	GB-WT	Relative Difference
Peak-to-valley (GW)	161.8 ± 38.9	194.3 ± 11.3	-16.7%
Minimum output ratio (%)	47.4 ± 8.1	29.9 ± 19.1	+58.5%
Ramping frequency (times/day)	52.1 ± 6.0	49.8 ± 8.7	+4.6%
Mean output (GW)	47.2 ± 17.2	92.6 ± 31.6	-49.0%

TABLE V
STATISTICAL CHARACTERISTICS OF WIND GENERATION FROM AWES AND GB-WT CONFIGURATIONS

Metric	AWES	GB-WT	Relative Difference
Mean (GW)	129.6 ± 16.4	83.8 ± 37.5	54.8%
Std. dev. (GW)	33.5 ± 12.6	56.1 ± 12.3	40.3%
Correlation with PV	-0.03 ± 0.15	0.12 ± 0.10	-

In addition, AWES exhibits slight negative correlation with PV generation, with a mean value of -0.03 ± 0.15 , indicating temporal complementarity between wind and solar. By contrast, GB-WT output shows a modest positive correlation with PV, 0.12 ± 0.10 , which limits potential synergy. The complementarity of AWES and PV allows more balanced renewable supply, further reducing reliance on storage and thermal flexibility, and enhancing overall system operational efficiency.

3) *BESS Dispatch Intensity*: To quantitatively evaluate the operational intensity of the BESS relative to its rated capacity, we introduce the Normalized Dispatch Range (NDR) indicator, defined as the ratio between the actual dispatch power range and the rated power capacity, i.e., $NDR = (P_{\max} - P_{\min}) / P_{\text{rated}}$, where $P_{\max}$ and $P_{\min}$ represent the maximum and minimum dispatch power during operation, and P_{rated} denotes the rated power capacity of the storage system.

As shown in Fig. 6, the NDR of AWES-related storage reaches 61%, whereas that of GB-WT storage is 48%. This implies that AWES storage utilizes more than half of its rated dispatchable capacity, operating closer to its power boundaries and exhibiting stronger power buffering characteristics. By

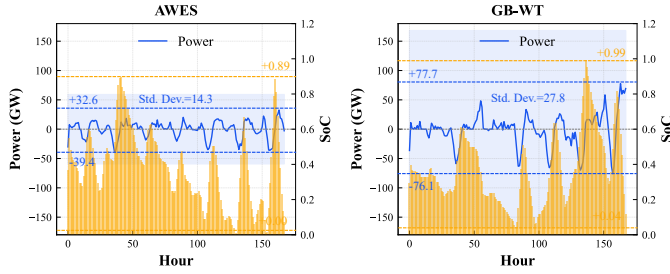


Fig. 6. Comparison of BESS operational behavior under AWES and GB-WT configurations.

contrast, GB-WT storage demonstrates relatively lower utilization, operating predominantly within the mid-range of its rated capacity and aligning more with an energy shifting role.

The comparison highlights that AWES storage is more engaged in short-term, fast-response regulation, directly contributing to system flexibility and volatility balancing. GB-WT storage, on the other hand, plays a greater role in inter-temporal energy management, such as peak shaving and load leveling.

G. Economic Assessment

Building on the capacity planning and operational results, Fig. 7 and Table VI present a comparison of the cost structures and the levelized cost of electricity (LCOE) for the AWES and GB-WT configurations. The AWES configuration achieves a total annualized cost of 457 billion CNY, which is 14% lower than the 533 billion CNY required for the GB-WT configuration. This cost advantage is primarily attributable to a 65% reduction in the required storage power capacity, and a 48% decrease in thermal generation. Although the installed wind capacity of the AWES configuration is only 18% higher than that of the GB-WT configuration, the resulting wind power generation is 61% greater than that of the GB-WT configuration. These results indicate a superior ability of AWES to capture high-altitude wind resources and to deliver a smoother temporal output profile.

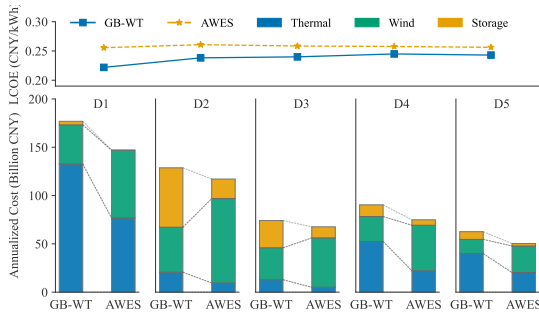


Fig. 7. Regional comparison of cost composition and LCOE.

Despite the overall cost reduction, the LCOE of the AWES configuration, at 0.258 CNY/kWh, is approximately 9.3% higher than that of the GB-WT configuration, which is 0.236 CNY/kWh. The higher LCOE is mainly due to increased capital investment in airborne wind infrastructure. While the costs associated with storage and thermal generation

TABLE VI
ECONOMIC INDICATORS OF AWES AND GB-WT CONFIGURATIONS

Metric	AWES	GB-WT
Annualized Cost Breakdown (Billion CNY)		
Storage-related cost value	40.0	114.0
Wind power-related cost	283.0	161.0
Thermal power-related cost	134.0	258.0
Total annualized cost	457.0	533.0
Electricity Generation and Demand (TWh)		
Thermal generation	446.0	861.0
Solar generation	291.0	283.0
Wind generation	1097.0	682.0
Load loss	0.0093	0.0023
Total demand	1825.0	1830.0
Total generation	1834.0	1827.0
Cost Efficiency (CNY/kWh)		
LCOE of Wind power generation technology	0.258	0.236

are substantially reduced, the savings are offset by the need to amortize the higher fixed costs over a comparable total electricity generation volume, namely 1834 TWh for AWES configuration and 1827 TWh for GB-WT configuration. Therefore, although AWES provides a more cost-effective system from a macroeconomic perspective, it incurs a slightly higher cost per unit of electricity produced.

H. Sensitivity Analysis

To evaluate the robustness of the results, the unit costs of storage and wind turbines are varied by $\pm 10\%$, $\pm 20\%$, and $\pm 30\%$ relative to their baseline values, as illustrated in Fig. 8.

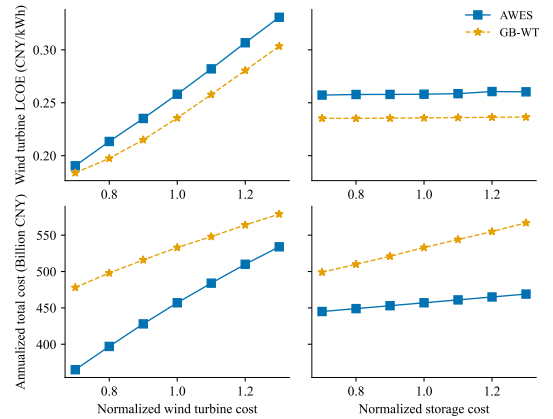


Fig. 8. Sensitivity of total annualized cost and LCOE to variations in storage and wind turbine unit costs.

The results reveal distinct sensitivities with respect to storage and wind turbine costs. Regarding storage costs, the AWES configuration demonstrates strong resilience: When the unit cost of storage increases by 30%, its total annualized cost rises from 357 billion CNY to 387 billion CNY, an increase of only 8.3%. Under the same condition, the LCOE remains within the range of 0.21 CNY/kWh to 0.28 CNY/kWh. By contrast, the GB-WT configuration experiences a 13.4% increase in total cost, from 496 billion CNY to 562 billion CNY, owing to its

storage requirements being approximately three to four times greater than those of the AWES configuration.

Turning to wind turbine costs, the AWES configuration is more sensitive due to its wind-dominated generation mix. An increase of 30% in the wind turbine unit cost leads to a rise in total annualized cost from 267 billion CNY to 385 billion CNY, corresponding to a 44.4% increase. The LCOE also increases from 0.19 CNY/kWh to 0.33 CNY/kWh, representing a 74% rise. In contrast, the GB-WT configuration is less affected, with total cost and LCOE increasing by 17.9% and 58%, respectively.

Overall, these results indicate that the AWES configuration attains its highest relative competitiveness in scenarios where storage costs are elevated, as its lower dependence on storage offsets its higher capital intensity. However, its economic performance remains vulnerable to increases in airborne wind turbine costs. From a long-term planning perspective, sustaining competitiveness will require targeted cost-reduction strategies for AWES technology, including improvements in design and manufacturing, the realization of economies of scale through large-scale deployment, and the establishment of supportive regulatory frameworks for airspace utilization.

VI. CONCLUSION

This paper develops an integrated framework combining altitude-based AWES optimization, two-stage stochastic planning, and problem-based scenario reduction to evaluate the techno-economic impacts of AWES deployment in multi-energy systems. The altitude optimization model enables dynamic adaptation to high-yield wind layers, improving wind capture and smoothing generation profiles. Embedded within a stochastic planning context, these operational benefits translate into significant system-level advantages, including up to 65% reduction in required storage capacity, smoother net load profiles, reduced cycling stress on storage and thermal assets, and approximately 14% lower total annualized system costs despite a slightly higher LCOE due to capital intensity. The *PCCK-Medoids* scenario reduction ensures computational tractability without sacrificing uncertainty representativeness, facilitating large-scale simulations. Sensitivity analyses demonstrate that AWES economic competitiveness is robust under high storage cost scenarios but sensitive to airborne turbine capital costs, highlighting the need for technological innovation and cost reduction. Overall, these results position AWES technology as a promising enabler for cleaner, more flexible, and cost-effective power systems, contingent on continued advancements in platform design and cost efficiency.

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